

CCNA Lab & Reference Manual

Multi-LAN Packet Tracer Topology, CLI Commands & Core Fundamentals

TOPIC

Cisco IOS CLI & Inter-VLAN Routing

TARGET CERTIFICATION

CCNA 200-301

ENVIRONMENT

Cisco Packet Tracer / 2960 & 2911

1. Complete Lab Topology Overview

This lab environment demonstrates multi-subnet routing across different local network segments using a central Cisco router (R1) connecting two switches and an isolated management host.

Device	Interface	Connected To	IP Address	Subnet Mask	Default Gateway	Cable Type
PC0	Fa0	Switch0 (Fa0/1)	192.168.1.10	255.255.255.0 (/24)	192.168.1.1	Straight-Through
PC1	Fa0	Switch0 (Fa0/2)	192.168.1.11	255.255.255.0 (/24)	192.168.1.1	Straight-Through
Switch0	Fa0/24	Router R1 (Gi0/0)	192.168.1.2 (SVI*)	255.255.255.0 (/24)	192.168.1.1	Straight-Through
PC2	Fa0	Router R1 (Gi0/1)	10.1.1.10	255.255.255.0 (/24)	10.1.1.1	Crossover
PC3	Fa0	Switch1 (Fa0/1)	172.16.1.10	255.255.255.0 (/24)	172.16.1.1	Straight-Through
PC4	Fa0	Switch1 (Fa0/2)	172.16.1.11	255.255.255.0 (/24)	172.16.1.1	Straight-Through
Switch1	Fa0/24	Router R1 (Gi0/2)	172.16.1.2 (SVI*)	255.255.255.0 (/24)	172.16.1.1	Straight-Through
R1 (Router)	Gi0/0	Switch0 (Fa0/24)	192.168.1.1	255.255.255.0 (/24)	N/A	Straight-Through
	Gi0/1	PC2 (Fa0)	10.1.1.1	255.255.255.0 (/24)	N/A	Crossover
	Gi0/2	Switch1 (Fa0/24)	172.16.1.1	255.255.255.0 (/24)	N/A	Straight-Through

* Note: SVI (Switch Virtual Interface) IPs are optional for switch management.

2. Essential Networking Rules & Conventions

Cabling Conventions (MDI vs. MDIX)

- **Copper Straight-Through:** Used for *unlike devices* (e.g., Switch to Router, Switch to PC). Switches use MDIX pinouts, while PCs and Routers use MDI pinouts.

- **Copper Crossover:** Used for *like devices* (e.g., PC to Router, Switch to Switch, PC to PC). Because both PCs and Router interfaces act as MDI end hosts, connecting a PC directly to a router requires a crossover cable.

Packet Tracer Auto-MDIX Alert

In Packet Tracer, crossover cables connected between a PC and Switch may turn green because **Auto-MDIX** is enabled on 2960 switch ports by default. However, for CCNA exam preparation, always select standard cabling conventions (Straight-Through for Switch-to-PC / Switch-to-Router).

IP Address Structuring & Subnet Masks

- **Default Gateway (.1):** First usable IP in a subnet is reserved for the router interface to allow hosts to locate their exit path easily.
- **Static Infrastructure (.2 – .9):** Reserved for Switch SVIs, wireless access points, network printers, and servers.
- **End-User DHCP Hosts (.10+):** Assigned to PCs, workstations, and mobile devices.
- **Classful vs. Classless (CIDR):** Class B addresses (e.g., 172.16.1.10) default to 255.255.0.0 in Packet Tracer. We override this to 255.255.255.0 (/24) so the host mask matches the router interface gateway.

3. Router & Switch CLI Configuration Scripts

Router R1 Initial Setup & Security

```
! Enter Configuration Mode and assign Hostname
enable
configure terminal
hostname R1

! Secure Privileged EXEC mode with encrypted password
enable secret Cisco123!

! Secure Console Line access
line console 0
password ConsolePass123!
login
exit
```

Interface IP Assignments & Activation

```
! Configure GigabitEthernet0/0 (LAN 1 Gateway)
interface GigabitEthernet0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
exit

! Configure GigabitEthernet0/1 (PC2 Direct Gateway)
interface GigabitEthernet0/1
ip address 10.1.1.1 255.255.255.0
no shutdown
exit

! Configure GigabitEthernet0/2 (LAN 2 Gateway)
interface GigabitEthernet0/2
ip address 172.16.1.1 255.255.255.0
no shutdown
exit
```

```
! Save configuration to NVRAM
end
copy running-config startup-config
```

4. Key Operational Concepts Explained

Interface Status Logs (%LINK & %LINEPROTO)

Executing `no shutdown` brings up the interface and generates system logs:

- **%LINK-3-UPDOWN:** ... state to up → **Layer 1 (Physical):** Electrical signals detected.
- **%LINEPROTO-5-UPDOWN:** ... state to up → **Layer 2 (Data Link):** Framing/keepalives established ("up/up").

Packet Routing vs. Frame Forwarding (Layer 3 vs. Layer 2)

When **PC0 (192.168.1.10)** pings **PC2 (10.1.1.10)** across subnets:

1. **PC0 (Layer 3 Check):** Compares destination IP `10.1.1.10` against its subnet mask. Determines it is remote and requires the Default Gateway (`192.168.1.1`).
2. **PC0 (ARP Lookup):** Checks ARP table for gateway IP `192.168.1.1`. Places the **Router R1 Gi0/0 MAC address** as the destination Layer 2 frame address, while keeping the destination Layer 3 IP as `10.1.1.10`.
3. **Switch0 (Layer 2 Switching):** Reads the destination MAC address and forwards the frame out port `Fa0/24` to R1.
4. **Router R1 (Layer 3 Packet Forwarding / Routing Engine):** Strips the incoming Layer 2 header, inspects the IP packet, checks its routing table, re-encapsulates the packet into a new Layer 2 frame, and transmits it out `Gi0/1` directly to PC2.